

Michigan Tech

# ODEs: Language of science

Motivation, Definitions etc.

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# Outline

- Why Differential Equations DEs dominate science and engineering.
- Three case studies: Gravity, Lorentz, and Rössler.
- What changed in the 1960s and 70s.
- Lots of examples Circuit Examples Sprott [1]

## Why Science and Engineering are all about DEs.

- Calculus is the language of change.
- Questions about how things change are DEs.
- For example, Newton's second law  $f = ma$  is a second order DE because the acceleration  $a = y''(t)$  is a second derivative.
- DEs come with Initial Conditions (ICs). For example,  $f = my''$  has an initial position  $y(0) = y_0$  and velocity  $y'(0) = v_0$ .
- An Initial Value Problem (IVP) is a DE with an appropriate set of ICs
- Interesting DEs almost all involve more than one function. This is an ODE system.

## Second-Order ODEs are common because of Newton $f = ma$ and Kirchoff

**Structural vibrations** A building is a coupled system of damped masses and springs.

Second order ODEs because of Newton  $f = ma$ .

**Electrical circuits** A complicated circuit is a coupled system of voltage differences. It is analyzed by breaking it down into simple loops. Kirchoff gives ODEs usually second order because the simplest passive circuit elements are

inductors  $\Delta v \approx q''$

resistors  $\Delta v \approx q'$

capacitors  $\Delta v \approx q$

# Kepler's Laws of Planetary Motion

- 1: **Elliptical orbits** Planets move on ellipses with the Sun at one focus.
- 2: **Equal areas** Equal areas are swept out in equal times.
- 3: **Harmonic law** Period squared  $T^2$  is proportional to the semi-major axis cubed  $a^3$ .

*Kepler's process:* Tycho Brahe's (with no telescope) Mars observations of Mars were good to (4-8 seconds) or roughly 1-2 arcminutes. Circular orbits missed Brahe's data by 8 arcminutes. The ellipse (with  $\epsilon_{Mars} \approx 0.093$  and  $\epsilon_{Earth} \approx 0.018$  )

$$r(\theta) = \frac{a(1 - \epsilon^2)}{1 + \epsilon \cos(\theta)}$$

## Gravity: Newton's Machinery

For  $n$  masses  $m_i$  at positions  $\vec{r}_i \in \mathbb{R}^3$  Newton's gravitation and second law give

$$m_i \vec{r}_i'' = - \sum_{\substack{j=1 \\ j \neq i}}^n G \frac{m_i m_j (\vec{r}_i - \vec{r}_j)}{\|\vec{r}_i - \vec{r}_j\|^3}, \quad \text{for } i = 1, \dots, n. \quad (1)$$

Canceling  $m_i$  gives the nonlinear ODE system

$$\vec{r}_i'' = -G \sum_{\substack{j=1 \\ j \neq i}}^n m_j \frac{\vec{r}_i - \vec{r}_j}{|\vec{r}_i - \vec{r}_j|^3} \quad (2)$$

This is the system that Poincaré was thinking about when he was developing the techniques and tools we are going to look at.

# Clockwork Vision

*The celestial machine is to be likened not to a divine organism but rather to a clockwork.*  
— Kepler 1605

*An intelligence which, at a given instant, knew all the forces by which nature is animated, and the respective positions of all the beings of which it is composed, and if it were moreover sufficiently vast to subject these data to analysis, would embrace in the same formula the motions of the greatest bodies of the universe and those of the lightest atom.*  
— Laplace, 1814

## Poincaré and Lorentz

*One will be struck by the complexity of this figure, which I do not even attempt to draw. Nothing can give us a better idea of the intricacy of the three-body problem, and of all the problems of dynamics in general ...a kind of lattice-work, a weave, a chain-link network of infinitely fine mesh.* — Poincaré 1899

*"Predictability: Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?"* — Lorentz 1972

*If the flap of a butterfly's wings can be instrumental in generating a tornado, it can equally well be instrumental in preventing a tornado.* — Lorentz 1972

# Lorentz

$$\begin{aligned}\dot{x} &= \sigma(y - x) & \sigma &= 10 \\ \dot{y} &= x(\rho - z) - y & \rho &= 28 \\ \dot{z} &= xy - \beta z & \beta &= \frac{8}{3}\end{aligned}$$

Lorentz observed chaos in this ODE model for a *highly* simplified weather system. This is the butterfly system common in non-technical discussions of chaos.

# Rössler

$$\begin{aligned}\dot{x} &= -y - z & a &= 0.2 \\ \dot{y} &= x + ay & b &= 0.2 \\ \dot{z} &= b + z(x - c) & c &= 5.7\end{aligned}$$

*I had no idea what he meant. - Rössler on being introduced to chaos 1975.*

*A prototype equation to the Lorenz model of turbulence contains just one (second-order) nonlinearity in one variable. - Rössler 1976*

*Simple systems can produce complicated behavior. The reason lies in the structure of space and time. Simple 'weaving rules' suffice to determine very complicated motions. - Rossler 2020*

# Rössler Chaos and Hyperchaos

**1976 Chaos: Std**  
 **$(a,b,c)=(0.2,0.2,5.7)$**

$$\begin{aligned}\dot{x} &= -y - z, \\ \dot{y} &= x + ay, \\ \dot{z} &= b + z(x - c).\end{aligned}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (+, 0, -)$$

**1979 Hyperchaos**  
 **$(a,b,c,d)=(0.25,3,0.5,0.05)$**

$$\begin{aligned}\dot{x} &= -y - z, \\ \dot{y} &= x + ay + w, \\ \dot{z} &= b + xz, \\ \dot{w} &= -cz + dw.\end{aligned}$$

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (+, +, 0, -)$$

We will learn what  $\lambda$  means and why the expanded 1979 version is *hyperchaotic*!

# Chaotic Jerk IVPS from Sprott

| No. | Jerk ODE                                                        | IC $(y, \dot{y}, \ddot{y})$ | Lyapunov exps      |
|-----|-----------------------------------------------------------------|-----------------------------|--------------------|
| 1   | $\ddot{y} = -2.017\ddot{y} \pm \dot{y}^2 - y$                   | $(0, 0, \pm 1)$             | $0.055, 0, -2.072$ |
| 2   | $\ddot{y} = -2.8\dot{y} \pm y + y^2$                            | $(\mp 0.5, -1, 1)$          | $0.002, 0, -0.002$ |
| 3   | $\ddot{y} = -0.44\ddot{y} - 2\dot{y} \pm (y^2 - 1)$             | $(0, 0, 0)$                 | $0.105, 0, -0.545$ |
| 4   | $\ddot{y} = -0.5\ddot{y} - \dot{y} \pm y \pm y^2$               | $(0, \pm 1, 0)$             | $0.094, 0, -0.594$ |
| 5   | $\ddot{y} = -2\dot{y} \pm ( y  - 1)$                            | $\pm(-1, -1, 1)$            | $0.003, 0, -0.003$ |
| 6   | $\ddot{y} = -0.6\ddot{y} - \dot{y} \pm ( y  - 1)$               | $(0, 0, 0)$                 | $0.036, 0, -0.636$ |
| 7   | $\ddot{y} = -0.3\ddot{y} - 0.3\dot{y} - D(y) + 1$               | $(0, 0, 0)$                 | $0.042, 0, -0.342$ |
| 8   | $\ddot{y} = -0.3\ddot{y} - 0.3\dot{y} - R(y) - 1$               | $(0, 0, 0)$                 | $0.042, 0, -0.342$ |
| 9   | $\ddot{y} = -2.9\dot{y} \pm (0.7y - D(y) + 1)$                  | $\pm(0, -0.5, 0.5)$         | $0.005, 0, -0.005$ |
| 10  | $\ddot{y} = -2.9\dot{y} \pm (0.7y - R(y) - 1)$                  | $\pm(0, -0.5, 0.5)$         | $0.005, 0, -0.005$ |
| 11  | $\ddot{y} = -0.5\ddot{y} - \dot{y} - y + \operatorname{sgn}(y)$ | $(0, 0, 0)$                 | $0.080, 0, -0.580$ |
| 12  | $\ddot{y} = -0.5\ddot{y} - \dot{y} + y - \operatorname{sgn}(y)$ | $(0, 0, 0)$                 | $0.080, 0, -0.580$ |

## Jerk IVPS from Sprott

| No. | Jerk ODE                                               | IC $(y, \dot{y}, \ddot{y})$ | Lyapunov exps    |
|-----|--------------------------------------------------------|-----------------------------|------------------|
| 13  | $\ddot{y} = -0.7\ddot{y} - \dot{y} - y + H(y)$         | $(0, 0, 0)$                 | 0.035, 0, -0.735 |
| 14  | $\ddot{y} = -0.4\ddot{y} - \dot{y} - y + 2S(y)$        | $(0, 0, 0)$                 | 0.050, 0, -0.450 |
| 15  | $\ddot{y} = -0.4\ddot{y} - \dot{y} + y - 2S(y)$        | $(0, 0, 0)$                 | 0.050, 0, -0.450 |
| 16  | $\ddot{y} = -0.19\ddot{y} - \dot{y} - y + 2 \tanh y$   | $(0, 0, 0)$                 | 0.100, 0, -0.290 |
| 17  | $\ddot{y} = -0.19\ddot{y} - \dot{y} + y - 2 \tanh y$   | $(0, 0, 0)$                 | 0.100, 0, -0.290 |
| 18  | $\ddot{y} = -3.7\dot{y} \pm (y - y^3)$                 | $(0, 0, 0)$                 | 0.020, 0, -0.020 |
| 19  | $\ddot{y} = -0.6\ddot{y} + 2.8\dot{y} - \dot{y}^3 - y$ | $(0, 0, 0)$                 | 0.110, 0, -0.710 |
| 20  | $\ddot{y} = -0.7\ddot{y} - \dot{y} + y - y^3$          | $(0, 0, 0)$                 | 0.100, 0, -0.800 |
| 21  | $\ddot{y} = -0.35\ddot{y} - \dot{y} - y + y^3$         | $(0, 0, 0)$                 | 0.100, 0, -0.450 |
| 22  | $\ddot{y} = -0.2\ddot{y} - \dot{y} \pm \sin y$         | $(0, 0, 0)$                 | 0.120, 0, -0.320 |

## Jerk ODEs from Sprott

$$\text{Diode } D(y) = \begin{cases} y & y \geq 0, \\ 0 & y < 0, \end{cases} = \max(y, 0)$$

$$\text{Relay } R(y) = \begin{cases} 1 & y \geq 0, \\ -1 & y < 0, \end{cases} = \text{sgn}(y)$$

$$\text{Saturation } S(y) = \begin{cases} -1 & y < -1, \\ y & |y| \leq 1, \\ 1 & y > 1, \end{cases} = \max\{-1, \min(y, 1)\}$$

## What changed

- From 1900 Poincaré developed tools and the awareness that for  $m \geq 3$  orbits were complicated.
- Starting in the 1960s and 70s computers made it possible for practitioners to solve ODE systems reasonably reliably and reasonably quickly.
- From the 1990s on software made it easy for anyone to solve ODE systems reliably and very quickly.
- More recently, additional software made visualization tools accessible.

## Summary

- ODEs are everywhere.
- Science thought *everything* ran like clockwork because "ODEs are everywhere!"
- Poincaré developed tools and terminology to understand orbital complexity.
- Chaotic behavior recognized as common once approximating ODEs was routine.

*Chaos: When the present determines the future, but the approximate present does not approximately determine the future* -Lorentz 1993

- [1] J. C. Sprott. "Simple chaotic systems and circuits". In: *American Journal of Physics* 68.8 (2000), pp. 758–763. DOI: 10.1119/1.19545.